

Performance Analysis of Fuzzy Logic Maximum Power Point Tracking Scheme for Solar PV System Under Varying Load and Atmospheric Conditions

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Abstract— Fuzzy logic control (FLC) is considered as one of the most effective algorithms for Maximum Power Point Tracking (MPPT) in Photovoltaic (PV) panels and this paper serves to investigate the performance of this algorithm. It describes the step by step modelling of a full PV system with detailed design of the FLC algorithm for the MPPT controller. The model is implemented in MATLAB/Simulink. The PV system with the FLC-based MPPT controller is subjected to uniform and non-uniform changes in weather conditions with an extensive list of test cases including partial shading condition (PSC) and load variation to cater for as many of the realistic weather changes that can happen during the usual operation of PV plants. The simulation results showed that FLC tracked the maximum power point (MPP) to an efficiency of 99% very quickly and with very small oscillations during step changes in irradiance and temperature. However, its performance decreased under non-uniform weather changes. FLC also showed very low efficiency during load variation.

Index Terms—Fuzzy logic control, MATLAB/Simulink, Maximum Power Point Tracking, performance analysis, solar photovoltaic system.

I. INTRODUCTION

Photovoltaic (PV) generation are known to have drawbacks such as low energy conversion efficiency, high starting cost and weather-dependent intermittent generation [1]. Since the efficiency of PV panels ranges between 15% and 21%, Maximum Power Point Tracking (MPPT) methods are needed to increase this efficiency by extracting maximum power during variation in weather conditions from PV panels.

The main aim of this paper is the performance analysis of a photovoltaic (PV) system, which includes a MPPT control system based on the fuzzy logic control (FLC) algorithm. Similar studies using the PV system modelled in MATLAB/Simulink can be found in [2], however, this paper investigates the performance of the FLC algorithm in tracking the MPP more extensively through additional test cases accounting for more realistic weather occurrences during normal operation of a PV system.

The paper is organized as follows: a step by step mathematical model of a PV system incorporating the FLC-based MPPT technique done in MATLAB/ Simulink is presented in Section II. Section III presents the results obtained from the performance analysis of the modelled PV system. Section IV concludes the paper and discusses future research ideas.

II. MATHEMATICAL MODELLING

A. Solar PV Modeling

The PV module “Eco Line 60/230-250W, LX-250P” [3] was chosen for implementation. A real module is used as its datasheet can be used to verify the accuracy of the mathematical model implemented in Simulink. The module consists of 60 cells in series with a total power capacity of 250 W under standard conditions (STC) that is atmospheric (module) temperature of 298 K (25 °C) and an irradiance of 1000 W/m². The electrical characteristics given in Table I are obtained from the manufacturer’s datasheet [3] and some from [1].

TABLE I. ELECTRICAL SPECIFICATIONS FOR PV MODULE USED IN MODELLING

Electrical Parameters	Value
Rated power P_{mpp} at STC	250 W
Rated voltage V_{mpp} at STC	30.75 V
Number of cells in series	60
Short-circuit current (I_{sc})	8.61 A
Open-circuit voltage (V_{oc})	37.41 V
Temperature coefficient of I_{sc} (K_i)	0.05 %/K or %/ °C
Temperature coefficient of V_{oc} (K_v)	−0.32 %/K or %/ °C
Series Resistance (R_s)	0.22 Ω
Shunt Resistance (R_p)	415 Ω
Diode ideality factor (A)	1.3

The single diode model shown in Fig. 1 was chosen for implementation. The motivation behind using this model is because the focus of this work is on the performance analysis of the FLC algorithm, thus, simplicity of the modelling of the PV module is an objective. Moreover, the adequacy of the single diode model is recognized in many literatures for simulation purposes.

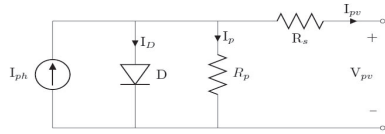


Fig. 1. Single PV cell equivalent model [1]

The cell photocurrent, I_{ph} , [2] can be calculated by:

$$I_{ph} = [I_{sc} + K_i(T_c - T_{ref})] \left(\frac{G}{G_{ref}} \right) \quad (1)$$

where I_{sc} is the short-circuit current [A], K_i is the temperature coefficient of the short-circuit current [%/K], T_c is the module temperature [K], G is the irradiation [W/m^2], $T_{ref} = 298$ K and $G_{ref} = 1000$ W/m^2

The reverse saturation current, I_{rs} , [2] is given by:

$$I_{rs} = \frac{[I_{sc} + K_i(T_c - T_{ref})]}{e^{\frac{q[V_{oc} + K_v(T_c - T_{ref})]}{N_s A k T_c}} - 1} \quad (2)$$

where K_v is the temperature coefficient of the open-circuit voltage [%/K], $k = 1.381 \times 10^{-23}$ J/K is the Boltzmann's constant and $q = 1.602 \times 10^{-19}$ C is the electron charge. The diode ideality factor A depends on the PV cell technology; e.g.: for polycrystalline silicon cells, $A = 1.3$.

Using Kirchhoff's Current Law, the output current, I_{pv} , of the PV module [1] in Fig. 1 is given by:

$$I_{pv} = N_p I_{ph} - N_p I_{rs} \left[e^{\frac{q \left(\frac{V_{pv} + I_{pv} R_s}{N_s} \right)}{A k T}} - 1 \right] - \frac{N_p V_{pv} + I_{pv} R_s}{R_p} \quad (3)$$

where V_{pv} is the output voltage of a cell [V], q is the electron charge (1.609×10^{-19} C), T is the cell absolute temperature [K], N_p is the number of cells in parallel and N_s is number of cells in series.

Equation (3) is modelled in Simulink environment using the basic mathematical operators. By doing so, it allows greater flexibility for variations of parameters of the PV module. Since the PV module chosen has no cells in parallel, the parameter N_p is omitted from equation (3).

Fig. 2 shows the model of the output current I_{pv} of the PV module as in equation (3) and Fig. 3 shows the subsystem of the modelled PV system implemented in Simulink. Notice that the PV module has both atmospheric temperature T_c ($^{\circ}C$) and irradiance G (W/m^2) as inputs which allows for different values of temperature and irradiance to be used to determine the characteristic curves of the PV module and in MPPT performance analysis. By modelling equation (3) we can simulate the circuit and extract close to real-world data of a PV module.

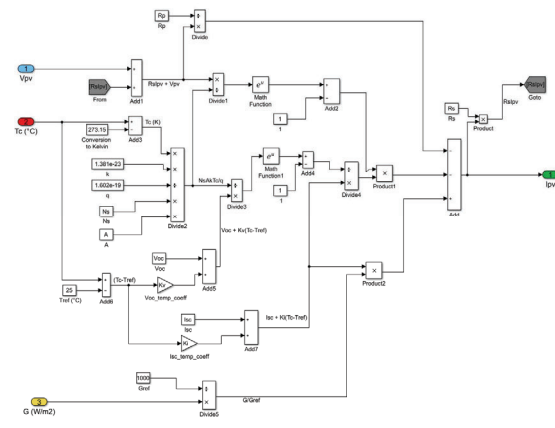
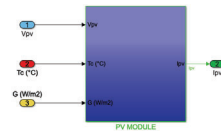

 Fig. 2. Full mathematical model of PV module output current I_{pv} .


Fig. 3. PV module subsystem developed in Simulink

B. DC-DC Boost Converter Model

As its name implies, this DC-DC converter boosts the output voltage from the PV module. It transfers maximum power from source to load by matching of source and load impedance. The MPPT controller adjusts the duty cycle of the transistor switch (a MOSFET in that case) by providing a duty ratio to the semiconductor which varies according to weather changes. The values for capacitance and inductance can be determined by using the standard equations for a simple boost converter as given in Table II.

 TABLE II. EQUATIONS TO CALCULATE MINIMUM INDUCTANCE L_{MIN} AND CAPACITANCE C OF BOOST CONVERTER FOR CONTINUOUS CONDUCTION OPERATION

OPERATION	
Minimum inductance, L_{min} based on ΔI_i	$L = V_d D / \Delta I_i f$
Capacitance, C	$C = \frac{D}{R \left(\frac{\Delta V_o}{V_o} \right) f}$

where, D is the duty cycle, V_o is the load voltage, [V], V_d is the input voltage that is same as the output voltage of PV module, [V], f is the switching frequency [Hz], R is the load resistance [Ω].

Table III shows the design specifications considered for the boost converter design.

TABLE III. DESIGN SPECIFICATIONS OF BOOST CONVERTER.

Specifications	Value
Input Voltage, V_d (V_{pv})	37.41 V
Output Voltage, V_o	74.82 V
Input Current, I_{pv}	8.61 A
Current Ripples, ΔI_i	1.722
Voltage Ripples, ΔV_o	10%
Switching frequency, f_s	1 kHz
Load resistance	10 Ω - 100 Ω

From the equations in Table II and the specifications in Table III, the value of the inductance and capacitance is calculated as $L = 10$ mH and $C = 50$ μF . The modelled circuit of the boost converter is shown in Fig. 4. The internal

resistance of both the MOSFET and the diode is reduced to 0.0001Ω to remove any additional resistance that they bring in the circuit and to reduce conduction losses especially during the operation of the MOSFET. The forward voltage drop of the diode is taken as 0.8 V to model a general-purpose diode.

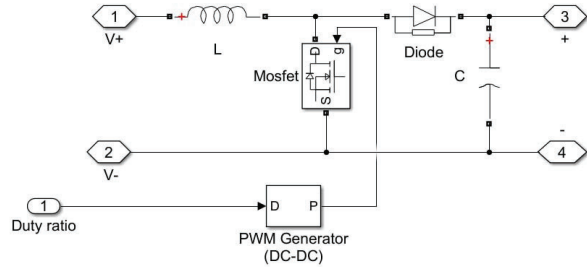


Fig. 4. Implemented circuit diagram of boost converter in Simulink

C. Fuzzy Logic Control (FLC) Algorithm

FLC is an attractive choice to implement MPPT in PV systems as it has the advantages of being independent of the PV system i.e. not requiring a mathematical model of the PV system and can handle the non-linearity associated with the P-V voltage characteristics of PV modules [4]. Fuzzy logic control generally consists of three stages: fuzzification, inference engine (rule base table lookup), and defuzzification as shown in Fig. 5 [5].

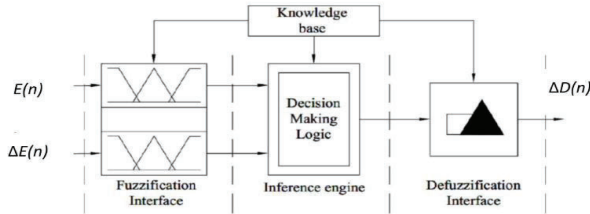


Fig. 5. General block diagram of fuzzy logic controller [5]

The FLC modelled accepts two inputs: error $E(n)$ and a change in error $\Delta E(n)$. Since at the MPP $E(n)$ is zero, these can be calculated using equation (4) and (5) as below:

$$E(n) = \frac{P(n) - P(n-1)}{V(n) - V(n-1)} \quad (4)$$

$$\Delta E(n) = E(n) - E(n-1) \quad (5)$$

Where, $P(n)$ and $V(n)$ are the instantaneous value of the power and voltage of the PV module modelled in Simulink. After calculating E and ΔE , the numerical value is converted to a linguistic variable based on the membership functions during the fuzzification stage

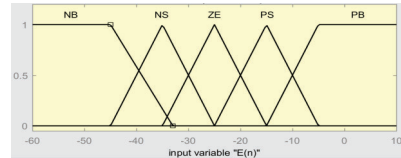
The linguistic variables of both $E(n)$ and $\Delta E(n)$ are as follows: NB (Negative Big), NS (Negative Small), ZE (Zero), PS (Positive Small), and PB (Positive Big). The duty ratio D uses the same linguistic variables as both $E(n)$ and $\Delta E(n)$. All the rules needed to determine the value of D is stored in the inference rule block which uses the 'Mamdani' method. There are 25 rules in all as shown in Table IV.

TABLE IV. LINGUISTICS VARIABLES OF MEMBERSHIP FUNCTIONS

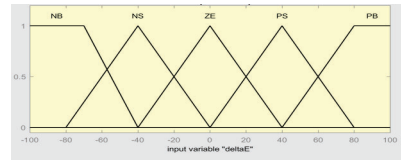
$\Delta E \backslash E$	NB	NS	ZE	PS	PB
NB	ZE	ZE	NB	NB	NB
NS	ZE	ZE	NS	NS	NS
ZE	NS	ZE	ZE	ZE	PS
PS	PS	PS	PS	ZE	ZE
PB	PB	PB	PB	ZE	ZE

The memberships function of input variables, error $E(n)$ and error change $\Delta E(n)$ and the output variable D are shown in Fig. 6. One of the major limitations of this method is defining the range of values for the error $E(n)$ and error change $\Delta E(n)$ as it requires the designer's experience with the system and may necessitates intense tuning of the memberships to achieve satisfactory performance in tracking the MPP. The range of the error $E(n)$ was deemed reasonable to be assumed to be from the negative value of the MPP to the positive value of the MPP at standard conditions as it can only vary between these two extremes. The change in error $\Delta E(n)$ was calculated by approximating the left and right side of the MPP on the P-V characteristics curve as straight lines and the positive and negative slopes calculated and taken as the range of values for $\Delta E(n)$.

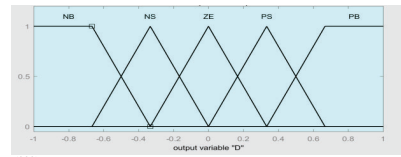
The estimated range of values for $E(n)$ and $\Delta E(n)$ worked only for certain irradiance level. To have the FLC working for most irradiance variation, some further tuning by trial and error change were done to the range for $E(n)$ and change in $\Delta E(n)$ until a satisfactory performance was obtained with the range of values for $E(n)$ and $\Delta E(n)$ as shown in Fig. 6. The triangular shape membership functions were found to work adequately and used throughout the modelling of the FLC.



(i)



(ii)



(iii)

Fig. 6. Triangular membership function for (i) error input variable $E(n)$, (ii), change in error input variable $\Delta E(n)$, (iii) duty ratio D

During defuzzification, the linguistic variable representing the duty ratio D is converted to a numerical value between 1 and -1 by using the "centroid" method which is the most commonly used method for defuzzification. The numerical value of D then goes through a saturation block which limits its value to between 0 and -1.

MATLAB/ Simulink provides a Fuzzy toolbox known as 'Fuzzy Logic Controller with Rule viewer' which allows for implementation of FLC for MPPT. The FLC works by providing the required duty ratio D needed to track the MPP to the gate of the MOSFET of the boost converter. The Simulink mathematical implementation of the FLC system is shown in Fig. 7 and its subsystem in Fig. 8.

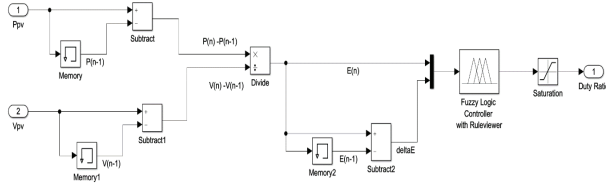


Fig. 7. Mathematical implementation of fuzzy logic controller system

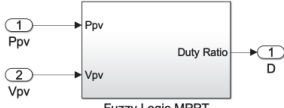


Fig. 8. Fuzzy logic controller subsystem

III. PERFORMANCE ANALYSIS AND DISCUSSION OF RESULTS

The performance of the FLC algorithms needs to be tested and analyzed. To do so, standard simulation scenarios were designed to answer the research questions as far as possible. The test cases are grouped into two categories, namely uniform weather conditions which includes uniform irradiance and uniform temperature change over the PV module and non-uniform weather conditions which consists of partial shading condition of a PV array. To simulate partial shading, three modules as designed in section II were connected in series to form an array. The main parameters used for performance comparison in all the test cases are maximum power tracked, the operating voltage, response/settling time and oscillations. The cases are explained below.

A. Test Case 1: Response of the FLC algorithm under uniform irradiance change

The FLC algorithm is simulated and analysed under the same variation of the level of irradiance while the atmospheric temperature is kept constant at 25°C throughout the full range of change in irradiance. The simulation time is 6s and consists of both linear and step variation in irradiance. This case simulates uniform cloud coverage of the PV module.

In order to show the response of the FLC algorithm to a sudden change in load, a circuit breaker is placed in parallel to the load and in series to an additional load (resistor) which creates an instantaneous change in resistance at $t=5s$. The current, voltage and power of both the PV module and boost converter at each irradiance is plotted against time. This is done to compare the output of the boost converter to that of the PV module to determine the efficiency of the boost converter and observe the response to the change in load.

Fig. 9 shows the irradiance pattern used to test the response of the fuzzy logic controller under uniform weather changes.

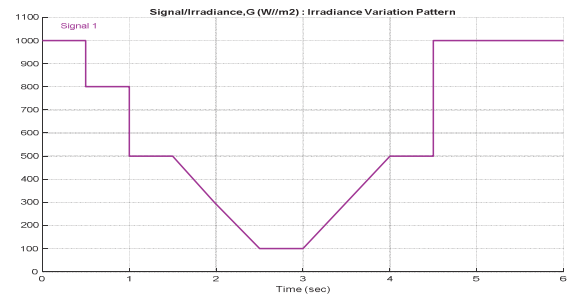


Fig. 9. Variation pattern of solar irradiance

Fig 10 shows the voltage, current and power coming out of the PV module which are the red lines and the blue lines are the outputs of the boost converter.

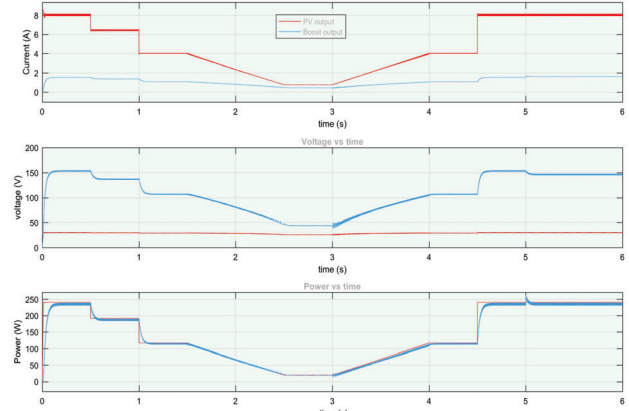


Fig. 10. Plots of output current, voltage and power of both the PV module (red) and boost converter (blue) against time for FLC system under uniformly varying irradiance

From Fig. 10, we observe that the FLC tracked the MPP during both step changes and ramp increase and decrease in irradiance. The FLC can reject the disturbance introduced at time $t=5s$ by the sudden change in load resistance as can be seen by the overshoot in the output power of the boost converter (blue line). It does the disturbance rejection in 0.02s. The FLC can keep the operation of the PV module at the MPP with an efficiency of 99% throughout the variation in irradiance as confirmed by the operating voltage being at V_{MPP} (30V) for the full duration of the simulation.

B. Test Case 2: Response of the FLC algorithm under partial shading condition (PSC)

To provide a thorough and rigorous performance analysis under PSC, the test case is broken down into four sub-cases 2A, 2B, 2C and 2D as follows:

B.1) Case 2A

The PV array is partially shaded at different irradiances with each of the three modules at a different irradiance. All the modules are at the same atmospheric temperature ($T=25^\circ\text{C}$). This setup creates three peaks in the P-V characteristic curve. The irradiances used are: PV1: 1000W/m², PV2: 800 W/m², PV3: 500W/m².

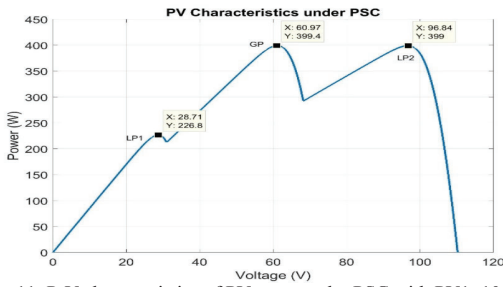


Fig. 11: P-V characteristics of PV array under PSC with PV1: 1000W/m², PV2: 1000 W/m², PV3: 800W/m²

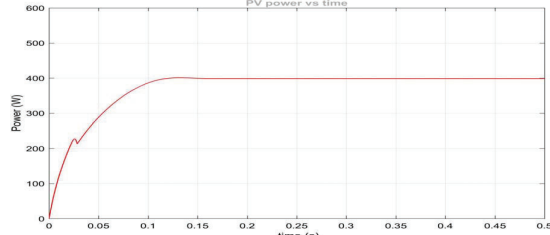


Fig. 12: Output power of the FLC algorithm under non-uniform irradiance of case 1 against time

From Fig. 12, the FLC controller tracks either the second local peak (LP2) or the global peak (GP) because they occur at almost the same power value but by looking at the voltage in Table V, we notice that the FLC is in fact tracking the GP which is usually not the case. The algorithm did a good job by passing through the first peak (LP1) without being trapped and moved on to track the GP. Table 6 also shows the performance values of the FLC-MPPT method. FLC is seen to respond and to settle very fast. It extracts maximum power at the GP and operates at V_{MPP} in steady state. FLC displayed very small variations in power output and can reasonably be ignored due to the very small values.

TABLE V. PERFORMANCE VALUES OF FLC MPPT (CASE 2A)

Parameters	FLC
Maximum Power (W)	399.1
Variations (W)	0.3
Maximum Voltage (V)	61.17
Settling time (s)	0.119
Efficiency (%)	99.9

B.2) Case 2B

In this case the dynamic behaviour of the FLC MPPT under PSC is tested with all the modules at the same atmospheric temperature ($T=25^{\circ}\text{C}$). This is done by varying the irradiance of each PV module in a different pattern at a specific time to create a dynamic change in irradiance. This case is designed to simulate as close as possible real-life variations in irradiance by passing clouds covering partially the PV array. The dynamic irradiance changes are given in Table VI.

TABLE VI. DYNAMIC IRRADIANCE CHANGES IN CASE 2B

No.	Irradiance Profile	Duration
SP1	(PV1: 1000 W/m ² , PV2: 800 W/m ² , PV3: 500 W/m ²)	0-1s
SP2	(PV1: 800 W/m ² , PV2: 500 W/m ² , PV3: 300 W/m ²)	1-2 s
SP3	(PV1: 500 W/m ² , PV2: 300 W/m ² , PV3: 100 W/m ²)	2-4S

Fig. 13 shows the P-V characteristics with all the local peaks and the global peak corresponding to each solar irradiance profile (SP1, SP2 and SP3) labelled. Note that since each PV modules is at a different irradiance, three

peaks are present. Fig. 14 shows the response and output power extracted by the FLC MPPT controller under dynamic non-uniform change in irradiance as described in Case 2B.

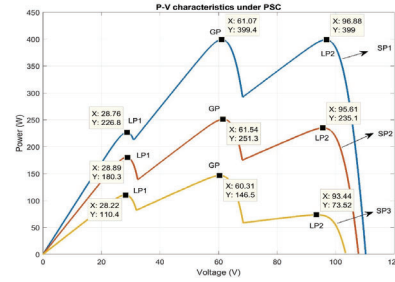


Fig. 13 P-V characteristics of PV array under dynamic non-uniform irradiance

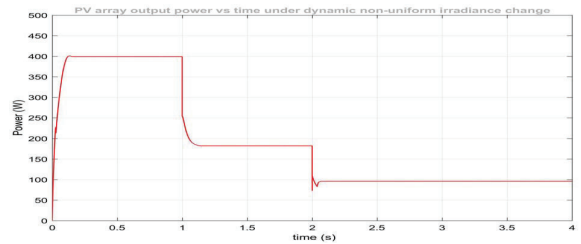


Fig. 14: The dynamic response during variation in irradiance pattern of FLC against time.

From Fig. 14, the fuzzy logic controller showed a very good tracking of one of the local peaks for each of the irradiance pattern described but an appreciable difference in power tracked is observed under the irradiance pattern SP3. FLC showed a small efficiency by tracking the second local peak (LP2) at 73 W from Fig. 13. Table VII shows the performance values of the FLC-MPPT method under dynamic irradiance change.

TABLE VII. COMPARISON OF MAXIMUM POWER EXTRACTED UNDER PSC DURING DYNAMIC IRRADIANCE CHANGE (CASE 2B)

Parameters	FLC		
	SP1	SP2	SP3
Maximum Power (W)	399.1	183.0	95.9
Variations (W)	0.13	0.02	0.03
Maximum Voltage (V)	61.17	42.75	30.85
Settling time (s)	0.263	0.255	0.074
Efficiency (%)	99.9	72.8	65.5

B.3) Case 2C: Dynamic Temperature Variation

In case 2C, the dynamic behaviour of all FLC algorithm under PSC with varying temperature is tested. This is done by changing the temperature of all the three modules simultaneously while keeping each of the three PV modules at a different irradiance such as shown below. This scenario is used to demonstrate the effect of temperature change during PSC. Table VIII shows the temperature and irradiance values used in Case 2C and Fig. 15 shows the response and output power extracted by the FLC MPPT controller under dynamic temperature change.

TABLE VIII. TEMPERATURE AND IRRADIANCE VALUES USED IN CASE 2C

Temperature	Solar irradiance profile	Duration
25°C	(PV1: 1000 W/m ² , PV2: 800 W/m ² , PV3: 500 W/m ²)	0-1s
10°C		1-2 s
45°C		2-4S

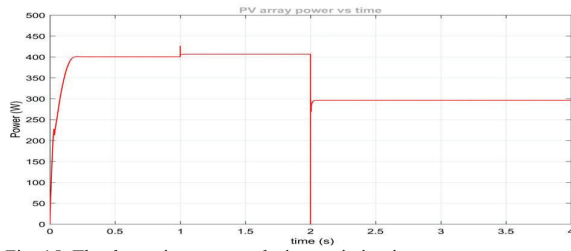


Fig. 15. The dynamic response during variation in temperature pattern of FLC against time

From Fig. 15 it is seen that the fuzzy logic controller was able to track the GP at temperature of 25°C and 10°C and showed fast responses. However, at 45°C, FLC operated in between the LP and the GP with a smaller efficiency than at the two other temperatures. An overshoot at time $t = 1$ s and a large undershoot at time $t = 2$ s are noticed in the FLC response. This can cause significant power loss in the PV system. Table IX compares the responses of each MPPT method under dynamic temperature change.

TABLE IX. COMPARISON OF MAXIMUM POWER EXTRACTED DURING DYNAMIC TEMPERATURE VARIATION (CASE 2C)

Temperature Parameters	FLC		
	25°C	10°C	40°C
Maximum Power (W)	401.1	450.8	282.9
Variations (W)	0.8	0.3	0.3
Maximum Voltage (V)	0.118	0.146	0.082

B.4) Case 2D: Variation of load

In case 2D, a load variation from 10 Ω to 100 Ω under PSC was performed to observe the response of the FLC MPPT. This scenario provides a realistic change that can happen during the usual operation of a PV plants. PV plants are very likely to be under PSC while a load change can happen at any time especially in residential PV plants. This combination is an important feature to analyse the performance of the FLC algorithm. The PSC condition applied to the PV array is as follows: (PV1: 1000 W/m², PV2: 800 W/m², PV3: 500 W/m²) with atmospheric temperature at 25°C. The response and the output power of the fuzzy logic-controlled system at maximum load of 100 Ω is shown in Fig. 16.

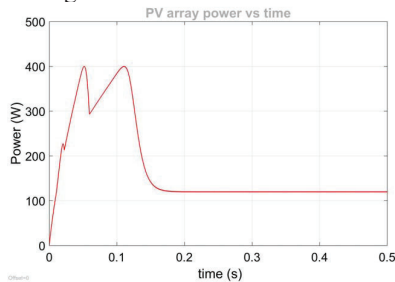


Fig. 16. The dynamic response of FLC against time during load change

The FLC response showed a very poor tracking behavior by operating well below the lowest local peak value. It was noticed through varying the load that the performance of FLC was greatly affected by load changes under PSC. The FLC had to be tuned after about every 5 Ω change in the load. In this case the FLC designed in Section II was left as it is without tuning to show that FLC needs constant tuning to perform adequately under changes of load and even under irradiance and temperature change as well. This shows the limitation and a significant drawback that FLC may have.

IV. CONCLUSION

This work describes the complete development of the model of a PV module using the single diode topology along with an FLC-based MPPT technique in MATLAB/Simulink software. The PV module and the MPPT controller modelled were subjected to a multitude of test cases consisting of uniform weather condition (irradiance and temperature) and non-uniform weather condition (PSC) including dynamic weather changes to analyze the performance of the FLC-based MPPT controller to such an extent that pertinent solutions can be provided to the research questions.

From the results and analysis in Section III, it can safely be said that a MPPT technique with the FLC algorithm is always a suitable option to guarantee MPPT both in fast changing and dynamically changing weather conditions. Not only has the FLC system modelled reached MPP accurately under uniform variation in irradiance and temperature and shown robustness against sudden load changes as in case 1, it has also demonstrated a very fast settling time which is important in limiting power loss. FLC was also able to track the local peak under PSC under most cases presented in this paper but showed very poor efficiency during load change under PSC. However, using FLC as a MPPT technique can come up with some challenges such as strong reliability on the designer's skills, requires significant amount of time in tuning the membership functions and error calculation, high complexity of implementation and the fact that it requires the specification data of the PV system to be implemented and most likely will not work on a different PV system.

A future research goal is implementing the FLC MPPT method simulated in this paper on a real PV module to control a boost converter with a load such as a DC motor to investigate further complexities and confirm the accuracy of the simulation reported in this paper.

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